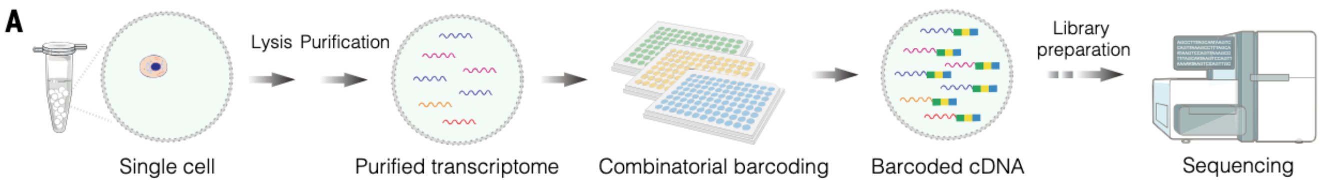
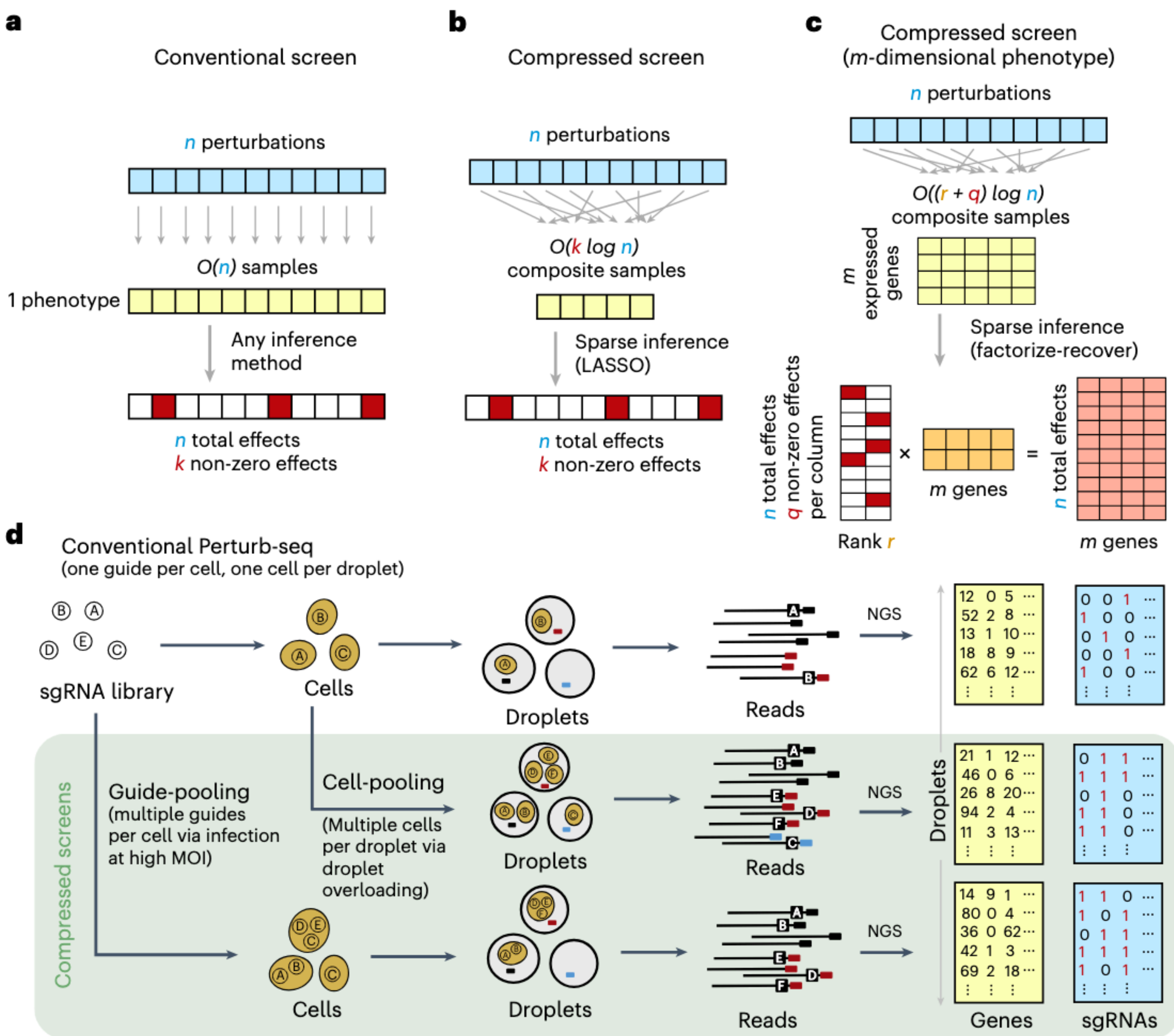
Pooled Industrial-Microbe CRISPRi with Clonal SPC-CapSeq

Reference Figures

CapSeq combinatorial barcoding – https://doi.org/10.1126/science.ady7227



Scalable screening by Perturb-seq – https://doi.org/10.1038/s41587-023-01964-9



Objective

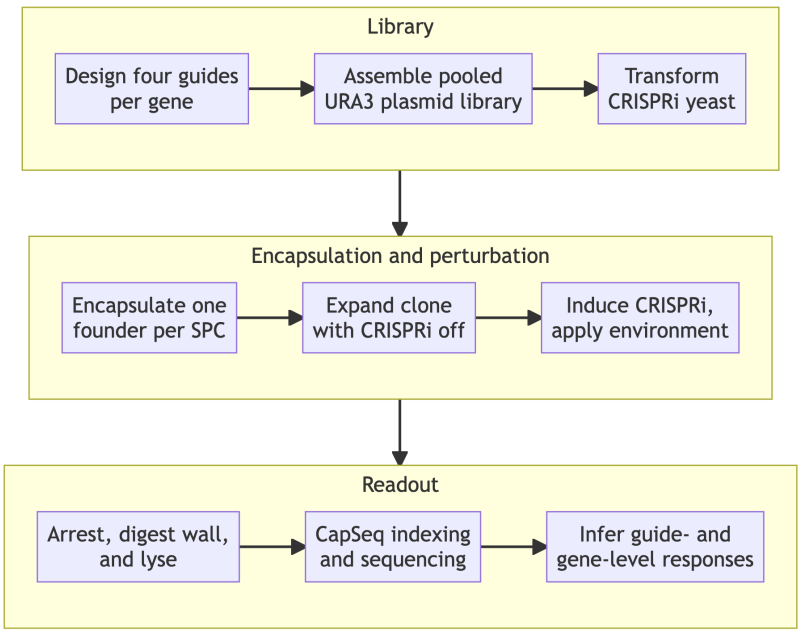
Build a pooled CRISPRi platform for **industrial microbes** in which one library cell is isolated per semipermeable capsule (SPC), expanded into a small clonal population, perturbed, and analyzed by capsule-resolved RNA sequencing. Clonal expansion supplies more RNA than a single microbial cell, while combinatorial CapSeq barcoding preserves the connection between a perturbation and its transcriptome.

The two primary hosts are *Saccharomyces cerevisiae* and *Pichia kudriavzevii*. *S. cerevisiae* is the chemistry-development chassis because its CRISPRi architecture, guide libraries, and reference transcriptome are already established; *P. kudriavzevii* is the industrial target of interest, valued for its acid, thermal, and osmotic tolerance and its use as an organic-acid and biofuel production host. The platform is designed so that the capsule workflow is shared and only the genetic parts differ between them. Other non-conventional yeasts and, with a changed capture chemistry, bacteria follow the same pattern. See Host Scope.

Each observation is

$$(b_c,\ g_c,\ e_c,\ \mathbf{y}_c),$$

where b_c is the capsule barcode, g_c is the CRISPRi guide, e_c is the environment (medium, drug, time point), and $\mathbf{y}_c$ is the clonal-population transcriptome. The environment index e_c is what turns a single screen into the experiment classes of Experiment Classes; the formal structure of the $\mathbf{y}_c$ is given in Data Model.



SPC Platform Capabilities

Every number below is from Baronas et al. 2026 (*Science* 391:1138, 10.1126/science.ady7227) unless marked otherwise. These are the measured properties the design must respect.

Property	Measured value	Design consequence
Capsule size	35–260 μm achievable; 70 $\pm$ 3.0 μm typical	Room for a microcolony; size is a tunable, not a fixed format
Shell thickness	2.0 μm (GelMA, photo-cross-linked)	Tunable via GelMA concentration
Nucleic acid retention	Complete for fragments $>$ 300 bp	Sets the minimum length of any RNA used as an identifier
Permeability	Enzymes and proteins up to $\approx$ 160 kDa enter; 500-kDa dextran does not	RT (71 kDa), Taq (97 kDa), and wall enzymes can be delivered
Pore size	$<$ 30 nm	Basis of the size cutoff
Mechanical resilience	Survives pipetting, centrifugation, freeze–thaw, thermocycling, solvents	Standard bench handling; no microfluidics after generation
Swelling	Up to 10$\times$ in volume without bursting as cells proliferate	Clonal expansion is accommodated by design

Property	Measured value	Design consequence
Microbial expansion	Bacteria and yeast expand into isogenic microcolonies	**Direct validation of the clonal-expansion premise for our hosts**
Cryopreservation	Capsules freeze and recover with minimal viability loss	A perturbed library can be banked mid-experiment
Cell release	Brief protease/collagenase treatment dissolves the shell, releasing viable cells	Hits can be recovered as live strains
FACS compatibility	Capsules sort on conventional instruments	Enables phenotype-first enrichment before sequencing
CapSeq depth	20,000 reads/cell $\rightarrow > 10{,}000$ UMIs, $> 3{,}000$ genes (K562); 88.85% mapping	Sets the read budget used throughout this note
Barcode space	$> 10^7$ combinations; $> 100{,}000$ cells per run at $\approx 1\%$ multiplet rate	Baseline; see 4-round extension

Two capabilities deserve emphasis because they change what experiments are possible, not merely how well they run.

Yeast and bacteria already expand into isogenic microcolonies inside SPCs. This is the single assumption our design most depends on, and it is reported rather than hypothesized. What remains unvalidated for us is not *whether* microcolonies form but whether the transcriptome survives wall digestion intact.

Capsules are sortable and cells are recoverable. Baronas et al. sorted capsules on an RNA marker by amplifying a target inside the capsule with fluorescent primers, then sequenced the enriched fraction. Combined with protease release of live cells, this closes a loop that a plate-based screen cannot: measure a transcriptome, select on it, and recover the strain that produced it.

Host Scope

The capsule workflow – generation, expansion, barcoding, sequencing – is host-independent. What changes between hosts is the genetic parts, the lysis chemistry, and whether oligo(dT) capture works at all.

Host	Class	Industrial interest	What changes
S. cerevisiae	Yeast	Reference chassis; ethanol, heterologous protein	Baseline: established dCas9-Mxi1 CRISPRi, genome-scale guide library, CEN/ARS + URA3
P. kudriavzevii	Non-conventional yeast	Organic acids, biofuels; acid, thermal, osmotic tolerance	Host-active promoters and markers; episomal replication is less reliable, so a broad-host ARS or genomic integration is needed; thicker/different wall may need a modified digestion
Y. lipolytica, *K. phaffii*	Non-conventional yeast	Lipids, secreted protein	Same class of substitutions as *P. kudriavzevii*
E. coli, *B. subtilis*	Bacteria	Chemicals, enzymes	**Capture chemistry changes fundamentally** – see below

The bacterial case is not a parts swap. Bacterial mRNA is not polyadenylated, so the oligo(dT) priming that CapSeq is built on captures nothing. A bacterial version requires either in-capsule polyadenylation before RT, random-primed capture with rRNA depletion, or a targeted probe panel. Two further differences compound this: rRNA is roughly 95% of bacterial total RNA, so undepleted libraries waste most of their depth; and bacterial mRNA half-lives are on the order of minutes, so the arrest step that is merely important in yeast becomes the dominant source of artifact. Bacteria should be treated as a separate development track, not a later milestone of the yeast one.

For the two primary yeasts, the *S. cerevisiae* arm develops and validates the chemistry, and the *P. kudriavzevii* arm inherits it. Running both through the same capsule workflow also gives a free and unusually clean control, described under Development Plan.

System Design

CRISPRi Host

Use a *ura3*$^{-}$ strain carrying genomically integrated *dCas9*-Mxi1 and TetR. The library plasmid supplies the guide and URA3 selection. This follows the established yeast CRISPRi architecture of an integrated effector and episomal, tet-inducible guides.

Induction is retained as experimental control, not as a requirement. The preferred workflow is to expand each founder for approximately five to seven doublings with CRISPRi off, then induce inside the SPC. This preserves guides against essential genes and prevents perturbation-dependent growth from eliminating the low-RNA advantage before measurement. A second arm can induce from the founder stage to measure chronic, growth-coupled responses.

Guide Library

Use four independent plasmids, each carrying one guide, per target gene. Select guides preferentially in the approximately 200-bp region immediately upstream of the transcription start site, with high chromatin accessibility, low nucleosome occupancy, and minimal predicted off-target activity.

The published genome-scale yeast CRISPRi library contains about ten guides for most genes. Its empirical rankings can be used to select four guides per gene rather than redesigning the entire library.

Plasmid Architecture

Use a CEN/ARS low-copy plasmid, not a 2μ vector:

$$\boxed{\text{CEN/ARS} + \text{URA3} + \text{inducible sgRNA} + \text{poly(A) perturbation-ID reporter}}$$

2μ vectors are high-copy and heterogeneous, commonly reaching tens of copies per cell. That variability would affect guide dosage, reporter counts, URA3 expression, cellular burden, and the measured transcriptome. CEN/ARS is the appropriate low-copy class, although it should not be assumed to be exactly one physical molecule per cell. The requirement is one plasmid genotype per founder; multiple identical copies are acceptable.

Perturbation Identification

The native sgRNA is only about 100 nt, well below the > 300 bp at which SPCs retain nucleic acids completely. It should not be the primary perturbation identifier. (The published retention curve is measured on DNA fragments, so the threshold for a structured single-stranded RNA should be confirmed empirically rather than assumed equal; the sgRNA is short enough that it fails either way.)

Instead, each plasmid should express a moderately abundant, polyadenylated reporter RNA approximately 700 to 1000 nt long. Place the guide's 20-nt target sequence, or a sequence-verified linked barcode, near the reporter's 3' end:

$$\text{moderate Pol II promoter} - \text{neutral RNA body} - \text{guide ID} - \text{poly(A) signal.}$$

The CapSeq oligo(dT) primer then captures both endogenous yeast mRNA and the perturbation-ID reporter. After capsule barcoding, the cDNA can be divided into a whole-transcriptome library and a small targeted perturbation-ID library. Capsules with two strong guide IDs are classified as multiplets.

Targeted amplification of retained plasmid DNA can be evaluated as an orthogonal backup, but it requires additional custom indexing chemistry.

Experiment Classes

The perturbation axis $\mathcal{L}$ and the environment axis $\mathcal{E}$ are independent, so the platform is not one screen but a family of them. A study is a choice of subset $\mathcal{L}' \times \mathcal{E}'$, and its capsule demand is $N = r\,|\mathcal{L}'|\,|\mathcal{E}'|$ for target replication r. Environments multiply capsule demand linearly, which is the main thing to budget for.

Media and Growth Conditions

The capsule shell is permeable to small molecules, so the medium can be changed after encapsulation without losing capsule identity. Capsules can also be split into aliquots after expansion and given different treatments, so one encapsulation supports several environments.

Conditions worth building the first panel from: carbon source (glucose, glycerol, ethanol, xylose, and for *K. phaffii* methanol); nitrogen limitation; low pH and organic-acid stress, which is the condition that makes *P. kudriavzevii* industrially interesting in the first place; elevated temperature; osmotic and ethanol stress. The readout is not growth but the transcriptome, so a knockdown that is neutral for fitness but shifts the stress program is still visible – which is precisely the class of gene that growth-based pooled screens miss.

Antifungal and Chemogenomic Targets

Dose a sub-inhibitory concentration of an antifungal (azoles, echinocandins, polyenes, or a candidate compound) as one of the environments. Each capsule then reports a transcriptome under a defined knockdown and a defined drug exposure, and the quantity of interest is the difference of differences

$$\varepsilon_{\ell j} \;=\; \theta_{\ell j}^{\mathrm{drug}} - \theta_{\ell j}^{\mathrm{ctrl}},$$

where $\theta_{\ell j}$ is the log fold change of gene j under element ℓ. A nonzero ε means the knockdown changes how the cell responds to the drug: chemogenomic interaction, in exactly the epistasis form this project already uses for genetic interactions. Sensitizers (targets whose knockdown potentiates the drug) and resistance determinants fall out of the same tensor.

Two readouts come free alongside the transcriptome. Capsule depth n_c is a biomass proxy, so a growth-based chemogenomic profile is recoverable from the same data (with the caveat that n_c also absorbs capture efficiency and needs a spike-in or a matched control to be trusted). And because capsules are FACS-sortable, a strong sensitizer can be enriched before sequencing rather than found afterward.

Fluorescent and Transcriptional Reporters

A GFP or other reporter cassette serves three distinct roles here, and it is worth keeping them separate.

1. **As a sequenced feature.** The reporter mRNA is polyadenylated, so CapSeq counts it like any other gene: it is simply an extra column of the matrix. Because it is read as sequence rather than light, reporters do **not** compete for spectral bandwidth – dozens of pathway reporters can be multiplexed in one strain, which no FACS-based design can do.
2. **As a sortable phenotype.** Fluorescent protein still enables capsule sorting and ONYX imaging, so a reporter can define a selected subpopulation for enrichment before the sequencing spend.
3. **As a calibration standard.** A constitutive reporter at known copy number gives a per-capsule internal control for capture efficiency, which is what makes the n_c biomass proxy above interpretable.

Reporters worth installing early: an unfolded-protein-response reporter (a *KAR2* or *HAC1* promoter fusion) for secretion-burden work, an oxidative-stress reporter, and a pathway-flux or titer proxy for whichever product the host is being engineered toward.

Selection and Strain Recovery

Because the shell dissolves under brief protease treatment and releases viable cells, a capsule identified as interesting can be reopened and its clone recovered. The screen therefore terminates in strains, not only in a data matrix. This is the step that turns a descriptive screen into a strain-design loop, and it is worth validating early even though it is not needed for the first production run.

Experimental Workflow

1. **Library construction.** Assemble pooled guide plasmids from an oligo pool, maintaining the guide and reporter ID in the same construct.
2. **Library validation.** Sequence after bacterial assembly and after yeast transformation to measure guide representation, dropout, abundance skew, and guide-ID linkage.
3. **Yeast transformation.** Transform the integrated CRISPRi host under noninducing conditions. Maintain at least 50- to 100-fold yeast transformant coverage per library element.
4. **Single-founder encapsulation.** Load the pooled library at low occupancy and use ONYX imaging to optimize singlet recovery. Multi-guide capsules are removed computationally.
5. **Clonal expansion.** Grow in selective SC-Ura medium for approximately five to seven doublings, producing roughly 32 to 128 cells per capsule.
6. **CRISPRi induction.** Add anhydrotetracycline through the SPC shell. Initial time points should distinguish a proximal response (approximately 3 h) from a more developed response (approximately 6 to 8 h).
7. **Transcriptome arrest and wall digestion.** Rapidly arrest the culture, add osmotic support, digest the cell wall with zymolyase or lyticase, lyse the resulting spheroplasts, and wash inhibitors through the capsule shell.
8. **CapSeq.** Add BC1 and a UMI during reverse transcription, followed by two split-pool ligation barcodes and a final sublibrary PCR index.
9. **Sequencing and analysis.** Assign reads to capsules, assign each capsule to a guide, reject multiplets, and estimate guide- and gene-level effects.

The wall-digestion interval is the largest biological risk because the cells may continue changing their transcriptome during spheroplasting. Matched conventional rapid-extraction controls are required to quantify this bias.

CapSeq Barcode Logic

A capsule receives a barcode trajectory

$$b_c = \mathrm{BC1}_i \| \mathrm{BC2}_j \| \mathrm{BC3}_k \| \mathrm{BC4}_l.$$

Published CapSeq uses three barcoding rounds − BC1 during reverse transcription, then BC2 and BC3 by split-pool ligation − followed by BC4 as a sublibrary index added during PCR. For R rounds of 96-well barcoding and S sublibrary indexes the barcode space is $B = 96^R \times S$, and for n capsules the approximate per-capsule collision probability is

$$P_{\text{collision}} \;\approx\; 1 - \exp\left(-\frac{n-1}{B}\right).$$

Evaluated at $n = 100{,}000$ capsules:

Design	Barcode space B	$P_{\text{collision}}$
96^3, no sublibrary index	884,736	10.7%
$96^3 \times 12$ (published CapSeq)	10,616,832	0.94%
96^4, no sublibrary index	84,934,656	0.12%
$96^4 \times 12$	1,019,215,872	0.01%

The published three-round design is right at the edge: it is what makes 100,000 capsules a natural run size, and it is why pushing past that number requires more barcode space rather than merely more reagent. **A fourth 96-well round lifts the ceiling by roughly two orders of magnitude.** At $96^4 \times 12$, one million capsules still collide at only

$$1 - \exp\left(-\frac{10^6 - 1}{1.019 \times 10^9}\right) \approx 0.1\%,$$

so barcode space stops being the binding constraint on run size entirely. This has a direct consequence for the genome-scale design, taken up under Scale.

The fourth round is not free. Each split-pool ligation loses material, so a fourth round trades barcode space against molecular recovery, and the loss should be measured in the pilot before it is committed to. It also adds a full 96-well plate cycle of handling per run.

Two different failure modes are both near 1%, and they should not be conflated. A *multiplet* is two founders physically sharing one capsule; it is set by the Poisson loading rate λ and is fixed by loading more dilutely. A *barcode collision* is two distinct capsules receiving the same barcode trajectory; it is set by B relative to n and is fixed by adding rounds. They are independent, and a design can be limited by either.

The capsule barcode identifies the physical clonal population; the reporter RNA identifies its perturbation. These are separate identifiers.

Data Model: The Matrix That Comes Out

Index Sets

$$\begin{array}{ll} c \in \mathcal{C} = \{1, \dots, N\} & \text{capsules (clonal populations) passing QC} \\ \ell \in \mathcal{L} = \{1, \dots, L\} & \text{library elements (guides and controls)} \\ e \in \mathcal{E} = \{1, \dots, E\} & \text{environments (medium, drug, time point)} \\ j \in \mathcal{J} = \{1, \dots, G\} & \text{measured features} \end{array}$$

The feature axis is the host transcriptome plus what we engineer into it:

$$G \;=\; \underbrace{G_{\text{endo}}}_{\text{host genes}} \;+\; \underbrace{G_{\text{rep}}}_{\text{reporters}} \;+\; \underbrace{1}_{\text{perturbation ID}}.$$

For *S. cerevisiae* $G_{\text{endo}} \approx 6{,}000$; *P. kudriavzevii* is similar at roughly 5,000 to 5,500; *E. coli* roughly 4,400. **This is the dimension of a single observation**: one capsule is a point in G-dimensional count space, so the transcriptome vector has about six thousand coordinates, not six thousand cells' worth of anything.

The Primary Observable and Its Domain

$$\mathbf{Y} = [y_{cj}] \in \mathbb{Z}_{\geq 0}^{N \times G}.$$

y_{cj} is the number of distinct mRNA molecules of gene j recovered from capsule c, after UMI deduplication. Three properties matter and are easy to get wrong:

- **The entries are non-negative integers, not positive reals.** They are molecule counts, so $\mathbb{Z}_{\geq 0}$ is the correct domain. Modeling them requires a count distribution – negative binomial, or Poisson with a gene-level overdispersion term – not a Gaussian.
- **Zero is common and ambiguous.** A zero is either structural (the gene is off) or sampling (the gene is on but was not captured at this depth). These are different events and no transformation distinguishes them.
- **The row sum is a nuisance parameter.** $n_c = \sum_j y_{cj}$ is the capsule depth, and it varies with clone size and capture efficiency, so raw rows are not comparable across capsules without normalization.

Only after processing do the values become real-valued, and the domain changes at each step:

$$\underbrace{\mathbf{Y} \in \mathbb{Z}_{\geq 0}^{N \times G}}_{\text{UMI counts}} \xrightarrow{\times 10^4 / n_c} \underbrace{\mathbb{R}_{\geq 0}^{N \times G}}_{\text{depth-normalized}} \xrightarrow{\log(1+\cdot)} \underbrace{\mathbb{R}_{\geq 0}^{N \times G}}_{\text{log scale}} \xrightarrow{\text{center}} \underbrace{\mathbb{R}^{N \times G}}_{\text{signed}}$$

So the answer to "are they positive reals" is: the measurement is a non-negative integer; the normalized expression value is a non-negative real; and only the effect size is a signed real. Effect sizes are the end product,

$$\Theta = [\theta_{\ell j}] \in \mathbb{R}^{L \times G}, \qquad \theta_{\ell j} = \log_2 \frac{\mu_{\ell j}}{\mu_{\text{ctrl},\, j}},$$

the log fold change of gene j under element ℓ against the non-targeting controls. These are genuinely all of $\mathbb{R}$: sign carries the direction of regulation.

Perturbation Assignment and Perturbations per Capsule

Assignment is a binary matrix $\mathbf{Z} \in \{0,1\}^{N \times L}$, where $z_{c\ell} = 1$ if capsule c carries element ℓ. In the intended design each capsule is clonal, so $\|\mathbf{z}_c\|_1 = 1$ exactly: **one perturbation per capsule, by construction.** Multi-guide capsules are a defect, not a feature.

The realized average follows from Poisson loading. If founders are loaded at mean occupancy λ, a capsule receives $k \sim \text{Poisson}(\lambda)$ founders, and the mean number of perturbations among *occupied* capsules is

$$\mathbb{E}[k \mid k \geq 1] \;=\; \frac{\lambda}{1 - e^{-\lambda}} \;\approx\; 1 + \frac{\lambda}{2}.$$

λ	Perturbations per occupied capsule	Singlet fraction	Capsules generated per usable singlet
0.05	1.025	97.5%	21.0
0.10	1.051	95.1%	11.1
0.20	1.103	90.3%	6.1

So the honest answer to "how many perturbations per capsule" is **just above one** – about 1.05 at a 10% loading rate – and the entire cost of keeping it there is generating roughly eleven capsules for every one you use. Since capsule generation is fast and cheap relative to sequencing, this is the right trade.

Replication is the complementary quantity:

$$r \;=\; \frac{N}{L\,E},$$

the mean number of capsules per (element, environment) cell of the design.

Size of the Final Matrices

Take the first production screen: $N = 100{,}000$ capsules, $L = 4{,}200$ elements, $G = 6{,}000$ features, one environment, at 20,000 reads per capsule (which Baronas et al. report yields > 10,000 UMIs and > 3,000 genes per cell).

Object	Shape	Entries	Domain	Size
Capsule × gene counts $\mathbf{Y}$	100,000 × 6,000	6.0×10^8	$\mathbb{Z}_{\geq 0}$	1.2 GB as `uint16`
Element pseudobulk $\mathbf{P}$	4,200 × 6,000	2.5×10^7	$\mathbb{Z}_{\geq 0}$	101 MB as `float32`
Gene-level effects Θ	1,000 × 6,000	6.0×10^6	$\mathbb{R}$	24 MB
Raw sequencing	2×10^9 read pairs	–	–	about 150–250 GB gzipped

Genome-scale changes only the leading dimension: Θ becomes 6,000 × 6,000, a **square gene-by-gene causal response matrix** of 3.6×10^7 coefficients at 144 MB. Adding environments multiplies through, giving a tensor $\Theta \in \mathbb{R}^{L \times E \times G}$.

Two observations about the capsule-level matrix that affect how it is stored. Expected occupancy is

$$\mathbb{E}[\#\{j : y_{cj} > 0\}] \;=\; \sum_{j=1}^{G} \left(1 - e^{-n_c p_j}\right),$$

with p_j the relative abundance of gene j; at $n_c \approx 10^4$ against $G \approx 6{,}000$ this should land near 40 to 70% density, far denser than mammalian scRNA-seq, because clonal expansion averages away the per-cell dropout that dominates single-cell data. The exact value must come from a downsampling curve, not from this formula's assumptions.

Consequently sparse CSR storage (about 1.8 GB at 50% density, carrying a 4-byte index per value) is *larger* than the 1.2 GB dense `uint16` array. **Store this matrix dense** – the usual single-cell sparse-matrix reflex is wrong here. `uint16` is safe because no single gene's count can exceed the capsule depth $n_c \approx 10^4$, well under the 65,535 ceiling.

What the Screen Yields

The measured quantity is one effect size per (perturbation, gene) pair, so a genome-scale run in one environment produces on the order of

$$L \times G \;\approx\; 24{,}500 \times 6{,}000 \;\approx\; 1.5 \times 10^8$$

guide-level coefficients, collapsing to 3.6×10^7 gene-level ones. This is the object that makes the screen worth running: not a list of hits, but a dense directed map from every perturbation to every transcriptional consequence, in a defined environment.

Scale

For T targeted genes, four guides per gene, and C control elements, the library size is

$$L = 4T + C,$$

and the capsule requirement for target replication r across E environments is $N = r\,L\,E$. (T is the number of genes *targeted*; G from the Data Model is the number of genes *measured*. At genome scale they coincide numerically, but they are different axes of the design.)

Target scope T	Elements L	Capsules at $r = 12$	Capsules at $r = 24$	Reads at $r = 12$
1,000 genes	about 4,200	50,400	100,800	1.0×10^9
2,000 genes	about 8,250	99,000	198,000	2.0×10^9
6,000 genes (genome-scale)	about 24,500	294,000	588,000	5.9×10^9

Genome Scale Fits in One Run with Four Barcoding Rounds

An earlier version of this design called for three separate 100,000-capsule runs at genome scale. **That recommendation was an artifact of the three-round barcode ceiling, and a fourth barcoding round removes it.** The reasoning is worth stating explicitly because it is the largest simplification available here:

- The 100,000-capsule run size is not a machine limit. It is where $96^3 \times 12$ barcode combinations put the collision rate at about 1%.
- At $96^4 \times 12 \approx 1.02 \times 10^9$ combinations, one million capsules collide at roughly 0.1%, so run size stops being barcode-limited.
- Genome-scale at $r = 12$ needs 294,000 capsules, comfortably inside that.
- Sequencing does not re-impose the limit: 5.9×10^9 read pairs fits a single high-output flow cell (a NovaSeq X 10B delivers roughly 10^{10}).
- Capsule generation does not re-impose it either: at 10% loading, 294,000 usable singlets require generating about 3.3 million capsules, under an hour of instrument time at either quoted generation rate.

So genome-scale becomes **one library, one run, one sequencing submission**, at the cost of one additional split-pool round. What must be verified in the pilot is the molecular yield lost to that fourth ligation, since that is the only term arguing the other way. Splitting into three runs remains the fallback if the fourth round proves lossy, but it should no longer be the plan of record.

Depth and Capsule Format

The 20,000 reads per capsule used throughout is calibrated to single mammalian cells and should not simply be inherited. A clonal population of 32 to 128 yeast cells has both more input RNA and a smaller transcriptome ($\approx$ 6,000 genes against $\approx$ 20,000), so its saturation point will differ, plausibly downward. Establish it by downsampling and saturation analysis in the pilot: per capsule depth multiplies straight through the largest cost term, so this is the single most valuable number the pilot can produce.

Baronas et al. report SPCs from 35 to 260 μm, with 70 $\pm$ 3.0 μm typical, that swell up to tenfold as cells proliferate. Larger capsules give a growing microcolony more room; smaller ones generate faster and consume less reagent. Choose the working format in the pilot by measuring singlet recovery and final colony size at two or three diameters. (Generation rates on the order of 2300 Hz for small capsules and 300 Hz for large ones have been quoted to us but do not appear in the published paper – confirm against a current Atrandi specification before they enter a budget.)

Cost Structure and Why the Library Should Be Built Full-Size Once

The three cost terms scale with entirely different quantities, and confusing them leads to a staging plan that spends more to do less.

Cost term	Scales with	Effect of library size L
Oligo pool synthesis	Per *pool*, weakly with member count	Essentially flat
Cloning, transformation, QC sequencing	Coverage $\times$ L	Mild
Split-pool barcoding (plates, enzymes, oligos)	Wells $\times$ rounds $\times$ runs	**None**
Sequencing	Capsules $\times$ depth	**None**

The barcoding and sequencing terms – which dominate – do not depend on library size at all. A 96-well round costs the same whether the pool contains 500 elements or 25,000, because you pay per well. **The marginal cost of screening a larger library at fixed capsule count is close to zero.**

This inverts the usual staging instinct. Synthesizing a small pilot library and later synthesizing a genome-scale one pays the synthesis and cloning setup twice, and the pilot's savings come out of the one term that was already cheap. The better structure is:

1. **Synthesize the genome-scale pool once**, with orthogonal primer-binding sites flanking defined subpools (by chromosome, by functional category, or by an arbitrary partition).
2. **Amplify out whichever subpool a given experiment needs.** A 500-element validation screen is then a PCR off the existing synthesis, not a new order.
3. **Choose experiment scope by how many capsules you process**, not by how many elements you cloned. Scope is a run-time decision, and $r = N/(LE)$ is the knob.

Two consequences follow. Sequencing depth per capsule is the term worth optimizing hardest, because it multiplies the largest cost by the largest count – which is why the saturation analysis above earns its place in the pilot. And underfilling a run is the main way to waste money: if the plates and the flow cell are committed, capsules left ungenerated are capacity paid for and not used.

These are structural claims about which costs scale with what, not a budget. Absolute figures need current quotes for oligo pool synthesis, CapSeq reagents, and sequencing before any of this becomes a number.

Wet-Lab Implementation

ONYX supplies controlled SPC generation, capsule-size monitoring, and occupancy optimization. Biofoundry liquid handling covers the 96-well distribution, pooling, redistribution, RT, ligations, sublibrary formation, cleanup, and QC – the split-pool rounds are exactly the kind of repetitive plate work that automation absorbs without added cost, which is what makes a fourth barcoding round cheap in labor terms.

ONYX and the SPC Innovator Kit do not by themselves provide a publicly listed turnkey RNA CapSeq workflow. Before purchase, confirm with Atrandi: access to the CapSeq oligos and barcode plates, the RNA chemistry itself, support for yeast wall digestion, whether a fourth split-pool round is supported by their barcode plate sets, and compatibility with a custom perturbation-ID reporter.

Development Plan

Pilot: Establish Chemistry

The pilot's purpose is to obtain **ground truth** – results the capsule workflow can be scored against, which a pooled screen by construction cannot provide, because in a pool you never independently know what was in a given capsule. Three deliberately different control designs supply it.

Arrayed known-genotype panel (10 to 20 strains). Build 10 to 20 strains individually rather than as a pool, each carrying a single sequence-verified guide, and process each strain through the capsule workflow *separately*. Because every capsule in a given batch has a known genotype, the sequencing result can be checked against a known answer. Choose targets whose knockdown phenotype is already published and easy to read out:

- three or four genes with large, well-documented transcriptional responses (for example *GAL4*, *HAP4*, *MSN2/4*), so the expected direction is known;
- two or three essential genes, to confirm the expand-then-induce schedule actually rescues guides that would otherwise drop out;
- two or three genes with no expected expression phenotype, as negative controls;
- several non-targeting guides, which become the θ denominator later.

Score three things per strain: does the intended target gene go down (the on-target check), does the rest of the transcriptome match bulk RNA-seq from the same strain grown conventionally (the fidelity check), and does the perturbation-ID reporter get recovered at the expected rate (the assignment check).

Species-mixing control. Co-encapsulate *S. cerevisiae* and *P. kudriavzevii*, then count capsules whose reads map to both genomes. Since the two species are unambiguous by sequence, this measures the combined multiplet-plus-collision rate empirically with no genetics required, and it calibrates the loading rate λ against a real number instead of a Poisson assumption. Running the two primary hosts as the mixing pair means the control also demonstrates that both hosts survive one shared workflow.

Guide-mixing control. Mix two strains carrying distinguishable perturbation-ID reporters at 1:1 and measure the fraction of capsules reporting both. This isolates multiplet rate specifically in the reporter channel that the real screen will depend on, which the species control cannot do.

Together these establish, in order of how likely each is to fail:

- transcriptome fidelity across wall digestion, against bulk RNA-seq;
- long-RNA retention during lysis and washing;
- wall-enzyme permeability and spheroplast formation in both yeasts;
- perturbation-ID recovery and correct capsule assignment;
- multiplet and barcode-collision rates, measured rather than assumed;
- single-founder loading, clonal expansion, and final colony size by capsule diameter;
- plasmid maintenance under SC-Ura selection through the expansion;
- the read-depth saturation curve, which sets the per-capsule budget;
- yield lost to a fourth split-pool barcoding round, if that round is adopted.

Small Pooled Validation

Amplify a defined subpool of approximately 100 genes with four guides per gene plus approximately 100 controls, giving about 500 elements, **out of the genome-scale synthesis** rather than ordering a separate small library. At 20,000 usable capsules this provides about 40 capsule replicates per element, enough to measure guide concordance and technical variance – and the subpool primer sites mean the same synthesis serves every later stage.

First Production Screen

Target 1,000 genes: approximately 4,200 elements. At $r = 24$ this is about 100,000 capsules and 2×10^9 reads. Well inside a three-round barcode design, so this stage can run before the fourth round is validated.

This is also the natural place to add the second axis: with $E = 2$ or 3 environments (a carbon-source shift, an antifungal at sub-MIC), the same 1,000 genes at $r = 12$ per condition needs 50,400 capsules per environment. Whether that fits one run depends on whether the fourth barcoding round is in place – which is the concrete reason to validate it during this stage rather than later.

Genome-Scale Screen

Target approximately 6,000 genes with four guides each plus controls, about 24,500 elements. **With four barcoding rounds this is one run**: 294,000 capsules at $r = 12$, roughly 5.9×10^9 read pairs, one flow cell. The three-run split is the fallback if the fourth ligation round costs too much yield, not the default plan.

Decision Gates

Proceed to scale-up only if the pilot demonstrates:

- reliable guide assignment for at least 90 to 95% of usable capsules;
- acceptable multi-guide and capsule-barcode collision rates;
- sufficient RNA and gene recovery after clonal expansion;
- limited transcriptome distortion from wall digestion;
- reproducible knockdown and concordant responses across guides;
- manageable guide-dependent differences in capsule population size;
- a saturation curve that fixes per-capsule read depth, since this sets the dominant cost term;
- for the four-round design specifically, molecular yield after the fourth split-pool ligation high enough to keep single-run genome scale worthwhile.

Open Questions

- **Wall digestion is the dominant biological risk.** Everything else has either been demonstrated in the published SPC work or is standard yeast genetics; the transcriptome's stability across spheroplasting has not been shown and is measured only by the bulk-RNA-seq comparison in the pilot.
- **Retention of structured single-stranded RNA** is extrapolated from a DNA-fragment retention curve. The reporter design has a wide margin above the 300 bp threshold, but the threshold itself should be confirmed for RNA.
- **CEN/ARS behavior in *P. kudriavzevii*** is not established the way it is in *S. cerevisiae*; a broad-host origin or genomic guide integration may be required, which would change the plasmid architecture for that host.
- **Bacterial capture chemistry** is an open development track, not a configuration change. Nothing in this note's poly(A) design carries over.
- **Vendor throughput and CapSeq kit availability** need confirmation in writing before a budget or purchase, as noted under Wet-Lab Implementation.

References

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- Scalable screening by Perturb-seq. *Nat. Biotechnol.* (2023). https://doi.org/10.1038/s41587-023-01964-9 (Pooled perturbation with transcriptomic readout at scale; the design pattern this note adapts to capsules and microbial hosts.)